

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – CONCEPT MAPPING

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL2, BL3, BL4)	Assessment:	Assessment Tool 3 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (5 Questions × 100 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
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Section / Batch:	CSE /2024	Date:	22/07/2026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2,Q3,Q4,Q5	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Concept Map 1: Decision Support Systems (DSS)

Task Description:

Construct a concept map that explains how a Decision Support System (DSS) supports organizational decision-making.

Core components: Data, Model, User Interface

1. Types of DSS
2. Flow: Data → Analysis → Decision
3. Role in business decision-making
4. Real-world example linkage

Concept Map 2: Operational Systems vs DSS Task

Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example systems (banking vs analytics)

Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

1. Data Sources → ETL → Data Warehouse → Data Marts
2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer

Concept Map 4: OLAP, Data Cubes & Dimensional Modelling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modelling techniques.

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

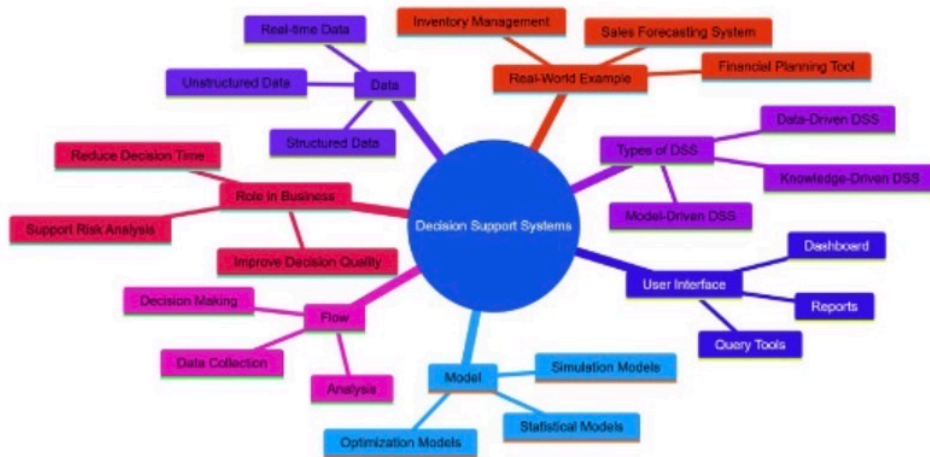
Concept Map 5: Query Processing & BI Tools (KNIME) Task

Description:

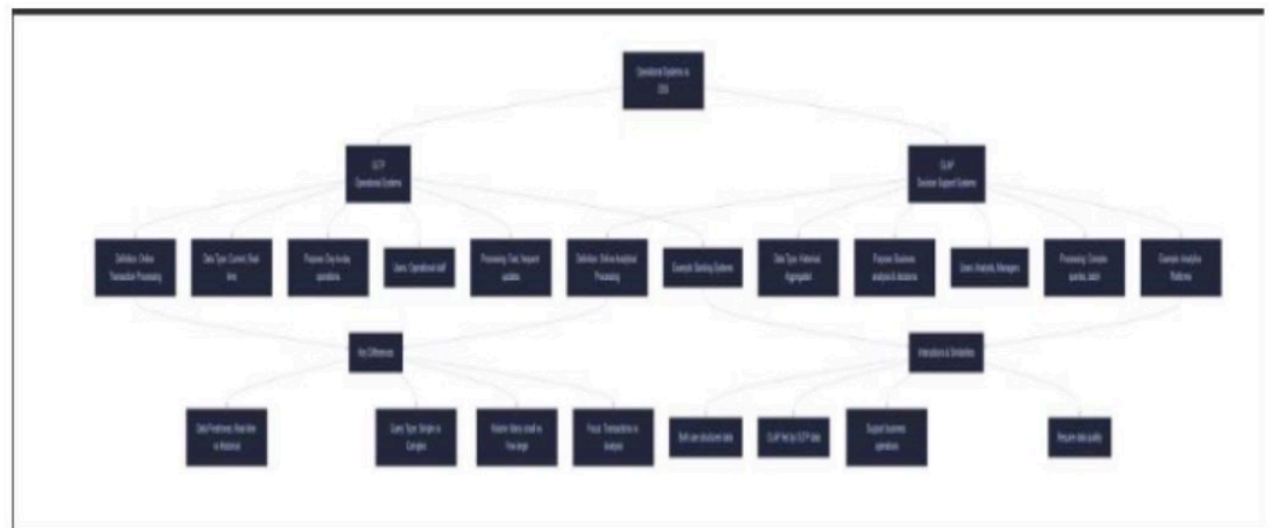
Construct a concept map explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

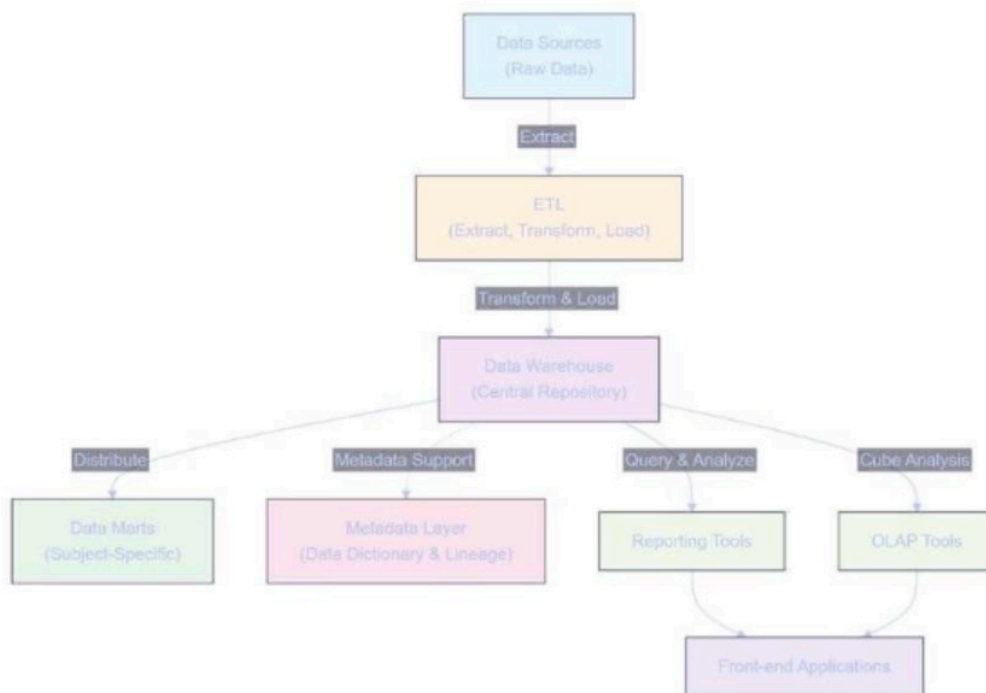
1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools

Concept Map 1: Decision Support Systems (DSS)

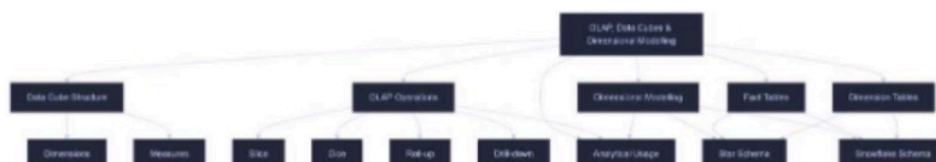


Concept Map 2: Operational Systems vs DSS Task





Concept Map 4: OLAP, Data Cubes & Dimensional Modelling



Concept Map 5: Query Processing & BI Tools (KNIME) Task

